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# Nutrition Fundamentals

*What to eat and how much: a free quick start*

FREE QUICK START

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FREE QUICK START

# Start here

This is the free quick start of Nutrition Fundamentals. It covers what matters most, how much to eat, protein, food quality and starter plans. The complete guide adds supplements, plateaus, diet breaks, eating out, ADHD and worksheets.

**The complete guide**

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CHAPTER 1

# What actually matters, in order

Think of nutrition as a pyramid. The bottom layers produce most of the result, and the top layers only matter once the bottom is solid. Most people spend their time at the top.

| Layer | What it is | How much it matters |
|---|---|---|
| 1 | **Consistency.** Can you actually follow it for months? | The biggest factor. Nothing works if you quit. |
| 2 | **Energy balance.** Calories in versus calories out | Decides whether you lose, gain or maintain weight. |
| 3 | **Protein.** Enough, spread across the day | Protects muscle, keeps you full, supports training. |
| 4 | **Food quality.** Mostly whole, minimally processed food | Affects hunger, health and how easy layer 2 is. |
| 5 | **Carbs and fat for your training.** Fuel for your sport | Matters more the harder and longer you train. |
| 6 | **Meal timing and frequency** | Small. Useful for convenience and hunger. |

| | | |
|---|---|---|
| 7 | **Supplements** | Smallest. A few have evidence, most do not. |

## Adherence beats the "perfect" diet

The strongest argument for starting with consistency comes from head-to-head trials. In a one-year randomized trial of 160 adults assigned to the Atkins, Zone, Weight Watchers or Ornish diets, **how much weight people lost was linked to how well they stuck with the diet (r = 0.60), not to which diet it was (r = 0.07).**[1] Adherence was low in every group, which is the real story: the diet that works is the one people keep doing.

The DIETFITS trial reached a similar conclusion with 609 adults randomized to a healthy low-fat or a healthy low-carbohydrate diet for 12 months. Average weight loss was 5.3 kg versus 6.0 kg, and the difference was not statistically significant. Neither genotype nor insulin response predicted who would do better on which diet.[2]

A broader look at tightly controlled feeding studies points the same way. A meta-analysis of 32 studies that swapped carbohydrate for fat at the same calories found only tiny differences: about 26 kcal per day more energy burned and about 16 g per day more fat lost on the **lower-fat** diets, the opposite of what low-carb advocates predicted.[3] Differences that small do not decide your results.

**What this means for you**

Stop searching for the best diet. Pick an approach that fits your food culture, budget and schedule, and put your effort into following it. Low carb, low fat, Mediterranean or flexible can all work if the calories and protein are right and you can stay with it.

CHAPTER 2

# Energy balance: how much to eat

To lose weight you need to take in less energy than you use. To gain you need more. This is not controversial, but it is **more dynamic than the textbook version.** When you eat less, your body burns somewhat less too, and your weight settles at a new level over many months rather than falling in a straight line.[4]

That is why a calculator gives you a **starting point and not an answer.** You will adjust it with your own data as you go.

## Step 1: estimate your maintenance

The Mifflin-St Jeor equation is the most reliable of the common formulas for estimating resting energy needs. A systematic review found it predicts within 10% of measured values more often than the alternatives, though it still misses for many individuals.[5][6]

| | Formula (kg and cm) |
|---|---|
| **Men** | (10 x weight) + (6.25 x height) - (5 x age) + 5 |
| **Women** | (10 x weight) + (6.25 x height) - (5 x age) - 161 |

Then multiply by an activity factor to estimate your total daily energy use. These multipliers are rough conventions, not measurements, so be honest and pick the **lower** one if you are unsure.

| Activity | Multiplier |
|---|---|
| **Mostly seated, little exercise** | x 1.2 |
| **Light exercise 1 to 3 days per week** | x 1.375 |
| **Moderate exercise 3 to 5 days per week** | x 1.55 |
| **Hard training 6 to 7 days per week** | x 1.725 |
| **Very hard training plus a physical job** | x 1.9 |

## Step 2: pick a deficit or a surplus

- **To lose fat:** eat about **300 to 500 kcal below** maintenance. A common target is losing roughly 0.5 to 1% of body weight per week, which lowers the risk of losing muscle.[6]
- **To build muscle:** eat about **200 to 300 kcal above** maintenance and train with resistance. Larger surpluses mostly add fat.
- **To maintain:** eat at your estimated maintenance and use the weekly check.

### The "3,500 calorie rule" is wrong

You may have heard that a 3,500 kcal deficit equals one pound of fat lost, so 500 kcal a day gives a pound a week. Dynamic models show this overestimates long-term loss, because your energy needs drop as you lose weight.[4] Expect fast early progress (partly water), then a slower, steadier rate. That is not a stall, it is physiology.

## A worked example

A 30 year old man, 80 kg, 175 cm, training four days a week. BMR = (10 × 80) + (6.25 × 175) - (5 x 30) + 5 = **1749 kcal**. Times 1.55 = about **2711 kcal** maintenance. For fat loss, subtract about 400: target roughly **2,300 kcal per day**.

**Your turn**

Write your own numbers in the worksheet at the end: BMR, maintenance, and your starting target. Round to the nearest 50. Do not agonize over the exact number, you will tune it.

CHAPTER 3

# Protein: the one macro worth tracking first

If you only track one thing, track protein. It builds and repairs muscle, it fills you up more than carbohydrate or fat, and it costs the body more energy to digest.[7][9]

## How much

A meta-analysis of 49 randomized trials with 1,863 participants found that the benefit of protein for gains in muscle plateaued at around **1.6 g per kg of body weight per day**, with the upper end of the confidence interval around 2.2 g per kg.[8] The ISSN position stand on protein and exercise lands in the same area for people who train, and calls 1.4 to 2.0 g per kg sufficient for most.[9]

| Situation | Daily protein (per kg body weight) |
|---|---|
| **General health, little training** | 0.8 to 1.2 g (a minimum, not an optimum) |
| **Training regularly, maintaining or gaining** | 1.6 to 2.0 g |
| **Fat loss while training** | 1.8 to 2.2 g |
| **Very lean athlete in a large deficit** | Possibly higher. A review suggests 2.3 to 3.1 g per kg of **lean mass**.[10] |

This is a range, not a cliff. The difference between 1.8 and 2.2 g per kg is small. Hitting 1.6 consistently beats hitting 2.4 occasionally.

## How to spread it

Eating protein at each meal is a practical way to reach your total and to control hunger. Research on muscle building suggests about **0.4 g per kg per meal** across at least four meals is a sound default, and that larger single doses are not wasted, just less efficient for muscle building at that moment.[11] For most people that means **25 to 45 g per meal.**

## Protein and hunger

In a controlled study, when 19 people raised protein from 15% to 30% of calories (keeping carbohydrate constant), they spontaneously ate about 440 kcal less per day and lost weight without being told to restrict.[12] Reviews of higher-protein diets point the same way for appetite and weight control.[13] It is a small study, but it is one of the more practical ways to make a deficit easier, and the main reason to put protein first.

| Food | Typical portion | Protein (approx.) |
|---|---|---|
| **Chicken breast, cooked** | 150 g | 45 g |
| **Eggs** | 3 large | 18 g |
| **Greek yogurt, plain** | 200 g | 20 g |
| **Canned tuna** | 1 can, drained (140 g) | 30 g |
| **Lean beef, cooked** | 150 g | 40 g |
| **Tofu, firm** | 150 g | 20 g |
| **Lentils, cooked** | 1 cup (200 g) | 18 g |
| **Whey protein** | 1 scoop (30 g) | 20 to 25 g |

Values are approximate and vary by brand and cut. Check the label or a food database.

CHAPTER 4

# Food quality and the 80/20 rule

Calories are the main driver of weight, but the **type of food changes how hard it is to control them.** The cleanest evidence comes from a tightly controlled trial at the US National Institutes of Health. Twenty adults lived in a metabolic ward and ate either an ultra-processed or an

unprocessed diet for two weeks each, with meals matched for presented calories, sugar, fat, fiber and sodium. They could eat as much as they wanted. On the ultra-processed diet they ate about **508 kcal more per day** and gained 0.9 kg. On the unprocessed diet they lost 0.9 kg.[14]

It is a small, short study, so do not read too much into the exact numbers. But it supports a simple practical idea: **build most of your meals from foods that look like the thing they came from**, because they are easier to stop eating.

## The 80/20 approach

- **About 80% of your food** comes from minimally processed sources: lean protein, vegetables, fruit, potatoes, rice, beans, oats, eggs, dairy, nuts and olive oil.
- **About 20% is for life:** the birthday cake, the pizza with friends, the thing you love. It counts toward your totals, and it does not count as failure.
- This is my coaching rule of thumb, not a measured threshold. What matters is the principle: **no food is banned.**

The reason to avoid bans has some evidence behind it. In a survey of 188 women, **rigid** dieting (all-or-nothing rules) was associated with more eating disorder symptoms, mood disturbance and higher BMI, while **flexible** dieting was not.[15] This was cross-sectional, so it cannot prove cause, but it matches what I see: people who treat one cookie as a disaster tend to turn it into a binge.

## A simple plate for most meals

1. **Protein** takes about a quarter of the plate (a palm or two).
2. **Vegetables or fruit** take about half.
3. **Carbohydrate** takes the rest, scaled to your training that day.
4. **Fat** is the cooking oil, the nuts, the cheese. Measure it at first, it adds up fast.

## Fiber and the micronutrient basics

Calories and protein get the attention, but a long series of reviews links **higher fiber and whole grain intake** with lower risk of heart disease, type 2 diabetes and colorectal cancer, and trials show lower body weight and cholesterol with higher fiber. The biggest benefit appeared at about **25 to 29 g of fiber per day**.[16] Most adults eat less. Beans, lentils, oats, vegetables, fruit, nuts and whole grains are the easy sources.

You do not need to count vitamins and minerals if you eat a varied diet with plenty of plants, some fish or eggs, and some dairy or fortified alternatives. Three cases deserve attention:

- **Vegan diets:** well planned vegetarian and vegan diets can be healthy, but need a reliable source of **vitamin B12** (fortified foods or a supplement), and attention to iron, calcium, iodine and omega-3.[17]
- **Heavy menstrual bleeding, frequent blood donation, or endurance training:** iron can run low. Test before supplementing, because too much iron is harmful.
- **Little sun exposure or darker skin at high latitudes:** vitamin D can run low. Ask your doctor to test it.

For anything beyond that, a blood test and a registered dietitian will tell you more than any generic list.

CHAPTER 5

# Pick your case: starter plans

| If this is you… | Your starter plan |
|---|---|
| **I want to lose fat** | Calculate maintenance, subtract 300 to 500 kcal. Protein 1.8 to 2.2 g per kg. Fat about 25%. Rest carbohydrate. Weekly check. Use the 80/20 plate. |
| **I want to gain muscle** | Maintenance plus 200 to 300 kcal. Protein 1.6 to 2.0 g per kg. Train with resistance. Weekly check, aim for 0.25 to 0.5% of body weight per week. |
| **I train hard (CrossFit, endurance)** | Raise carbohydrate with training load. Protein 1.6 to 2.0 g per kg. Eat before and after hard sessions. |
| **I have ADHD** | Default meals, meal alarms, track protein only. Do not track everything on day one. |
| **I yo-yo, binge, or hate being strict** | Skip the calories. Do the food quality chapter only (80/20, no banned foods) plus a protein target for four weeks. Talk to a professional if it feels out of control. |
| **I am a beginner and overwhelmed** | One change: protein at breakfast. Add one more change after 30 days. Read the free TRANSFORM quick start ("one thing at a time"). |

Want more? The complete guide has every chapter, worksheets and the full reference list. adrianonutrition.com/guides

**Questions?**

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